

EXPERIENCE

Jacobs Engineering

Guaynabo, PR

Software Engineer — Part-Time Foundry Developer

Jan 2026 – Present

- Architected an end-to-end AI resume transformation pipeline in Palantir Foundry — ingesting PDF/DOCX resumes up to 23 pages / 60K+ characters, extracting structured data via Claude Sonnet, scrubbing PII, enriching from HR datasets, and auto-generating formatted Word documents — cutting processing time from 4+ hours to ~2 minutes (98% reduction).
- Built a large-scale prompt engineering system using few-shot examples and chunked processing to support 4 resume formats and extract 65+ projects from a single document; deployed to production with automated scheduling across 30+ commits.
- Developed an ML water-tank forecasting platform for PRASA's infrastructure digitization initiative — processing 22M+ SCADA sensor records across 1,107 tanks using PySpark and AutoGluon — achieving a 56% improvement over persistence baseline (RMSE: 2.336 → 1.865) with probabilistic quantile outputs deployed to operational dashboards.

Engineered 30+ predictive time-series features, identified and corrected training-data contamination, and built a residual analysis validation framework for regional model assessment.

Software Engineering Intern

May 2025 – Aug 2025

- Built a workforce PDF automation tool that bulk-split large employee timesheet documents into individual submissions and packaged it as a GUI executable for non-technical staff — reducing a full day of manual work to ~5 minutes.
- Developed a Python ETL and reporting engine for CIP project timeline analytics — ingesting multi-format CSVs, normalizing milestone data, parsing varied date formats, and generating Excel dashboards with forecast vs. actual comparisons — recognized by senior leadership for improving project visibility across infrastructure programs.
- Built an Oracle-integrated Palantir Foundry dashboard for tracking approved vs. worked employee hours, enabling resource reallocation and real-time management oversight.

Software Engineering Intern

Jun 2024 – Aug 2024

- Designed and built a Palantir Foundry data response application for a private water utility — architecting ingestion, transformation, and validation pipelines for raw SCADA operational data.
- Developed an NLP-powered PDF chatbot to surface relevant internal procedures on demand, improving compliance accessibility and reducing lookup time for non-technical staff.
- Collaborated with senior engineers on QA testing, dataset validation, and performance optimization across Foundry pipelines.

Nagnoi LLC

San Juan, PR

Data & Software Engineering Intern

May 2023 – Jul 2023

- Partnered with the Puerto Rico Department of Education to build dashboards in Power BI tracking KPIs for school resource allocation across the island — supporting data-driven policy decisions for district administrators.
- Wrote SQL queries for data retrieval, cleaning, and statistical reporting; contributed to peer code reviews to maintain deliverable quality across the analytics team.
- Built Excel-based reporting views to present school performance and resource utilization metrics to non-technical stakeholders.

PROJECTS

OnboardOS — Healthcare Workflow Platform (Medido Health Capstone)

Next.js · Go · PostgreSQL · REST

- Co-built a full-stack healthcare onboarding platform with a workflow engine, role-based access control, audit logging, task lifecycle management, dependency resolution, and approval flows — shipping 40+ REST APIs with automated testing and cloud deployment.
- Owned dashboard development, analytics visualizations, notification systems, and workflow monitoring UI across the full-stack product.

Local RAG Knowledge Assistant

Ollama · Docker · RAG · Local LLM

- Designed and deployed a fully local retrieval-augmented generation system — document ingestion, embedding-based retrieval, and LLM inference via Ollama — containerized with Docker for reproducible, zero-external-API operation across environments.
- Exposed the pipeline through a REST API tested with Postman, enabling structured Q&A over private document collections and validating the full retrieval-to-generation flow end-to-end.

TECHNICAL SKILLS

Languages

Python, SQL, TypeScript, JavaScript, Java, C++, Go

AI / ML

LLM APIs (Claude, Ollama), Prompt Engineering, RAG, AutoGluon, scikit-learn, TensorFlow, Time-Series Forecasting

Data

PySpark, Pandas, Palantir Foundry, PostgreSQL, MySQL, Oracle, Power BI

Web / Backend

React, Next.js, Node.js, NestJS, Go, REST APIs, Tailwind CSS

Tools

Docker, Git, Linux, Jupyter — AI dev: Cursor, Claude Code, GitHub Copilot

EDUCATION

University of Puerto Rico – Mayagüez (UPRM)

Aug 2020 – May 2026

B.S. in Software Engineering · Minor Certificate in Project Management

Relevant Courses: Machine Learning Algorithms, Software Engineering Requirements, Database Systems, High-Performance Computing, Analysis of Algorithms, Software Reliability & Testing, Operating Systems, Computer Networks